

WORKSHEET**Percentage problems**

- 1 Mohammed got 54 correct answers out of 60 in his driving knowledge test. What percentage was this?
- 2 A soccer team won 72% of 75 games played. How many games did they win?
- 3 Brown's store offers a 15% discount while Wong's store offers a 16% discount. Whose store offers the better discount?
- 4 In a national survey of 12 270 citizens, 39% wanted a new Australian flag. How many people was this?
- 5 Increasing an amount by 5% is equivalent to multiplying that amount by 105%. What percentage is equivalent to:
 - a an increase of 12%?
 - b a decrease of 16%?
 - c a decrease of 9%?
- 6 Last year, 6 420 510 Australians had internet access but this year the figure increased by 9%. What is the new figure?
- 7 Kayla spent \$730 on her credit card and had to pay 13.49% interest. How much interest was this?
- 8 In a survey of 7400 people, 2812 of them approved of stricter gun laws. What percentage was this?
- 9 Will pays 28% of his weekly income in tax. If he earns \$844 per week, find his weekly tax.
- 10 The value of Steve's car depreciated by 18% this year. If its value last year was \$21 065, what is its value now?
- 11 At Westvale High, 48 of the 75 teachers are women. What percentage of the teachers are men?

- 12** There are 204 students in Year 8. If 88% of them went on a school camp:
- a** how many students went?
 - b** what percentage stayed home?
- 13** In one year, Australia's population of 19 450 200 increased by 1.1%. What was the new population?
- 14** At the office, 17% of the 82 employees smoke. How many people smoke?
- 15** What is the price of a mobile phone marked at \$109 after GST of 10% is added?
- 16** In Canada, a \$172 camera attracts GST of \$12.04. What is the percentage rate of GST?
- 17** So far, Nala has saved \$3085 towards a holiday trip and this is 62% of the full amount. What is the full amount for the trip? Answer correct to the nearest dollar.
- 18** Last weekend, 8400 drivers were breath-tested at random.
- a** If 105 drivers were charged, what percentage of the total tested were charged?
 - b** If 16% of the charged drivers were women, how many men were charged?
- 19** Calculate the sale price of a scooter marked at \$89 after a 15% discount is given.
- 20** The most popular Australian pet is the bird, making up 34% of all pets. If there are 20 470 100 pets in Australia, how many are birds?
- 21** A discount of \$50.40 was given on a CD stereo with a marked price of \$280.
- a** Find the discount price.
 - b** Calculate the discount as a percentage of the marked price.
- 22** Natasha earns 2.8% commission on all property she sells. How much did she earn for selling an apartment for \$185 750?

- 23** Emad's salary of \$40 200 received an increase of 4.6% this year. Calculate his new salary.
- 24** In a music survey of 1400 people, 22% preferred country music.
- a** What percentage didn't prefer country music?
 - b** How many people didn't prefer country music?
- 25** J-Mart are having a '12% off' sale. How much should Nicky pay for a tennis racquet marked at \$74?
- 26** In a survey, 430 people wanted Australia to become a republic, 870 people wanted Australia to remain a monarchy, while 100 were undecided. What percentage (to one decimal place) of the group wanted a republic?

Answers

- 1** 90%
- 2** 54
- 3** Wong's (16 %)
- 4** 4785
- 5** **a** 112% **b** 84% **c** 91%
- 6** 6 998 356
- 7** \$98.48
- 8** 38%
- 9** \$236.32
- 10** \$17 273.30
- 11** 36%
- 12** **a** About 180 students **b** 12%
- 13** 19 664 152
- 14** 14 people
- 15** \$119.90
- 16** 7%
- 17** \$4976
- 18** **a** 1.25% **b** 88
- 19** \$75.65
- 20** 6 959 834
- 21** **a** \$229.60 **b** 18%
- 22** \$5201
- 23** \$42 049.20
- 24** **a** 78% **b** 1092
- 25** \$65.12
- 26** 30.7%